Bank Database Schema:

```
create type account_type_typ as object (
 code char(4),
 name varchar2(50),
 interest float,
 min_bal float
);

create type account_typ as object (
 account_no char(10),
 account_ty ref account_type_typ,
 balance float
);

create type account_tlb as table of account_typ;

create table account_types of account_type_typ (code primary key);

create type customer_typ as object (
 id char(10),
 name varchar2(50),
 accounts account_tlb
);

create table customers of customer_typ(id primary key)
 nested table accounts store as ntlb_accounts;

insert into account_types values ('SAVS', 'General Savings', 5.24, 100.0);
```

```
insert into account_types values ('CHEQ', 'General Checking', 0.0, 0);
insert into account_types values ('HAPS', 'Hapan Savings', 6.26, 20000.0);

insert into customers values (
 '781250401V',
 'Sampath Weerasinghe',
 account_tlb(
  account_typ('1122300004',
        (select ref(s) from account_types s where s.code = 'SAVS'), 5700.00),
  account_typ('2334453124',
        (select ref(s) from account_types s where s.code = 'CHEQ'), 2300.00)
 )
);

insert into customers values (
 customer_typ (
  '99334453V', 'Dulani Pieris',
  account_tlb(
   account_typ('3445663321',
        (select ref(s) from account_types s where s.code = 'HAPS'), 19045.06),
   account_typ('5000022123',
        (select ref(s) from account_types s where s.code = 'SAVS'), 55235.00)
  )
 )
);
```

Question 1:

Deposit Rs. 1200/- to the account (Acc # 1122300004) of Mr. Sampath Weerasinghe (ID # 781250401V).

To deposit Rs. 1200/- to the account (Acc # 1122300004) of Mr. Sampath Weerasinghe (ID # 781250401V), you would need to update the balance of the respective account in the database. Here's the SQL statement to perform the deposit:

```
UPDATE TABLE(SELECT c.accounts FROM customers c WHERE c.id = '781250401V') a

SET a.balance = a.balance + 1200

WHERE a.account_no = '1122300004';
```

This SQL statement first selects the accounts belonging to Mr. Sampath Weerasinghe, identified by their ID. Then, it updates the balance of the specific account (Acc # 1122300004) by adding Rs. 1200 to the current balance.

Question 2:

Write a member function (called TotBal) that returns the total balance of a customer's accounts.

To create a member function called `TotBal` that returns the total balance of a customer's accounts, you can define it within the `customer_typ` object type. Here's how you can do it:

```
CREATE OR REPLACE TYPE customer_typ AS OBJECT (

  id CHAR(10),

  name VARCHAR2(50),

  accounts account_tlb,
```

-- Member function to calculate total balance

```
  MEMBER FUNCTION TotBal RETURN FLOAT
);
```

-- Body of the member function

```
CREATE OR REPLACE TYPE BODY customer_typ AS
  MEMBER FUNCTION TotBal RETURN FLOAT IS
    total_balance FLOAT := 0;
  BEGIN
    FOR i IN 1..self.accounts.count LOOP
      total_balance := total_balance + self.accounts(i).balance;
    END LOOP;
    RETURN total_balance;
  END;
END;
/
```

In this code:

- We define a member function `TotBal` within the `customer_typ` object type. This member function iterates through each account of the customer and calculates the total balance by summing up the balances of all accounts.

- `self` refers to the current instance of the `customer_typ`.

- `self.accounts(i).balance` accesses the balance attribute of each account within the accounts table of the customer.

Now, you can use this member function to retrieve the total balance of a customer's accounts. For example:

```sql
SELECT c.id, c.name, c.TotBal() AS total_balance
FROM customers c;
```

-This query will return the ID, name, and total balance of each customer.

Question 3:

Write a query to print the names of customers and their total balances (using the function created in question 2).

```sql
SELECT c.name AS customer_name, c.TotBal() AS total_balance
FROM customers c;
```